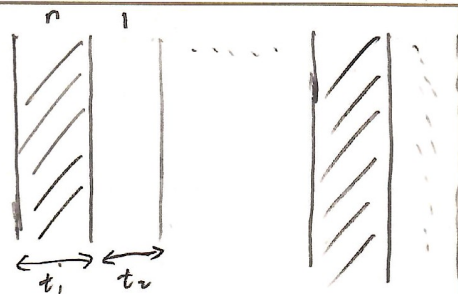


7.8

(a) Consider a unit consisting of a medium layer and an air gap, the transfer matrix for this unit is

$$T = \frac{1}{2} \begin{pmatrix} 1 + \frac{1}{n} & 1 - \frac{1}{n} \\ 1 - \frac{1}{n} & 1 + \frac{1}{n} \end{pmatrix} \begin{pmatrix} e^{ik_1 t_1} & \\ & e^{-ik_1 t_1} \end{pmatrix}$$

$$\times \frac{1}{2} \begin{pmatrix} 1+n & 1-n \\ 1-n & 1+n \end{pmatrix} \begin{pmatrix} e^{ik_2 t_2} & \\ & e^{-ik_2 t_2} \end{pmatrix}, \quad (k_1 = n\omega/c, \quad k_2 = \omega/c)$$



$$= \frac{1}{4n} [I(n+1) + \sigma_1(n-1)] [I \cos(k_1 t_1) + i\sigma_3 \sin(k_1 t_1)]$$

$$[I(1+n) + \sigma_1(1-n)] [I \cos(k_2 t_2) + i\sigma_3 \sin(k_2 t_2)]$$

$$= \frac{1}{4n} [I(n+1) \cos \alpha_1 + \sigma_1(n-1) \cos \alpha_1 + i\sigma_3(n+1) \sin \alpha_1 + \sigma_2(n-1) \sin \alpha_1] \quad \left(\begin{array}{l} \alpha_1 = n\omega t_1/c \\ \alpha_2 = \omega t_2/c \end{array} \right)$$

$$[I(1+n) \cos \alpha_2 + \sigma_1(1-n) \cos \alpha_2 + i\sigma_3(1+n) \sin \alpha_2 + \sigma_2(1-n) \sin \alpha_2]$$

$$= \frac{1}{4n} [I(n+1)^2 \cos \alpha_1 \cos \alpha_2 - \sigma_1(n-1) \cos \alpha_1 \cos \alpha_2 + i\sigma_3(n+1) \cos \alpha_1 \sin \alpha_2 - \sigma_2(n-1) \cos \alpha_1 \sin \alpha_2$$

$$+ \sigma_1(n^2-1) \cos \alpha_1 \cos \alpha_2 - [I(n-1)^2 \cos \alpha_1 \cos \alpha_2 + \sigma_2(n-1) \cos \alpha_1 \sin \alpha_2 - i\sigma_3(n-1) \cos \alpha_1 \sin \alpha_2$$

$$+ i\sigma_3(n+1) \sin \alpha_1 \cos \alpha_2 + \sigma_2(n^2-1) \sin \alpha_1 \cos \alpha_2 - I(n+1) \sin \alpha_1 \sin \alpha_2 - \sigma_1(n^2-1) \sin \alpha_1 \sin \alpha_2$$

$$+ \sigma_2(n^2-1) \sin \alpha_1 \cos \alpha_2 + i\sigma_3(n-1) \sin \alpha_1 \cos \alpha_2 - \sigma_1(n-1) \sin \alpha_1 \sin \alpha_2 - I(n-1) \sin \alpha_1 \sin \alpha_2]$$

$$= \frac{1}{4n} [I \{ (n+1)^2 \cos(\alpha_1 + \alpha_2) - (n-1)^2 \cos(\alpha_1 - \alpha_2) \} \\ + i\sigma_3 \{ (n+1)^2 \sin(\alpha_1 + \alpha_2) + (n-1)^2 \sin(\alpha_1 - \alpha_2) \} \\ - 2\sigma_1(n^2-1) \sin \alpha_1 \sin \alpha_2 + 2\sigma_2(n^2-1) \sin \alpha_1 \cos \alpha_2]$$

We can think of the stack as having $N-1$ material and air gap layers, with a final layer of material, which can be viewed as a film layer with the wave propagated back.

$$\text{Then, } T_{\text{stack}} = T^N (I \cos \alpha_2 - i\sigma_3 \sin \alpha_2).$$

(b) If both the air gaps and the foils have optical thicknesses of a quarter wavelength then $d_1 = d_2 = \pi/2$, and

$$T = \frac{1}{4n} \left[(n+1)^2 \cos \pi - (n-1)^2 \cos 0 \right] I + 2\sigma_1 \sin\left(\frac{\pi}{2}\right) \sin\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2} \left[\left(n + \frac{1}{n}\right) I - \left(n - \frac{1}{n}\right) \sigma_1 \right] = - \left[I \cosh \lambda - \sigma_1 \sinh \lambda \right]$$

where $\lambda = \log n$. Notice that

$$\exp\{-\lambda \sigma_1\} = \sum_{n=0}^{\infty} \frac{(-\lambda \sigma_1)^n}{n!} = I \sum_{n=0}^{\infty} \frac{\lambda^{2n}}{(2n)!} - \sigma_1 \sum_{n=0}^{\infty} \frac{\lambda^{2n+1}}{(2n+1)!} = I \cosh \lambda - \sigma_1 \sinh \lambda.$$

then we can write T as

$$T = - \exp\{-\lambda \sigma_1\}.$$

Then the transfer matrix for the stack is

$$T' = T^N \left(I \cos\left(\frac{\pi}{2}\right) - i \sigma_3 \sin\left(\frac{\pi}{2}\right) \right) = (-)^N \begin{pmatrix} \cosh(N\lambda) & -\sinh(N\lambda) \\ -\sinh(N\lambda) & \cosh(N\lambda) \end{pmatrix} \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$= (-)^N \begin{pmatrix} -i \cosh(N\lambda) & -i \sinh(N\lambda) \\ i \sinh(N\lambda) & i \cosh(N\lambda) \end{pmatrix}$$

From 7.8(c), we know

$$\left| \frac{E_{\text{trans}}}{E_{\text{inc}}} \right| = \left| \frac{\det T'}{t_{22}'} \right| = \frac{\cosh^2(N\lambda) - \sinh^2(N\lambda)}{\cosh(N\lambda)} = \text{sech}(N\lambda),$$

$$\text{and } \left| \frac{E_{\text{trans}}}{E_{\text{inc}}} \right|^2 = \text{sech}^2(N \log n) = \frac{4}{(e^{N \log n} + e^{-N \log n})^2} = \frac{4n^{2N}}{(n^{2N} + 1)^2}.$$

For $n^{2N} \gg 1$, the above result becomes asymptotically

$$4n^{-2N} = 4 \exp\{-N \log(n^2)\}$$